

Thomas Gooding, PhD

Research Scientist | Data Analyst | Medical Writer

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Professional summary:

- Biomedical research scientist with subject matter expertise in sleep physiology, exercise science, sports medicine, and human performance; bringing a rigorous scientific foundation to health technology and analytics roles.
- Proficient in Python and SQL for data wrangling, statistical modeling, and visualization; experienced in analyzing physiological and biometric datasets across large-scale and longitudinal research contexts with the ability to rapidly adopt new tools to meet evolving team needs.
- Collaborative, cross-functional researcher with a proven ability to design studies, interpret complex datasets, and communicate findings clearly to both scientific and non-technical stakeholders.

Work Experience:

Data Analyst | Affiliate Research Scientist

Feb 2026 – Present

Rutgers University, New Brunswick, NJ

- Querying and analyzing a pre-existing dataset of 1,300+ participants using MySQL to extract and process heart rate variability (HRV), respiratory data and related psychophysiological metrics across a broad range of physiological and demographic variables; producing data visualizations to communicate findings to scientific and applied audiences.
- Analyzing psychophysiological and allostatic load data in collegiate-aged populations using Python (VS Code, Jupyter Notebook; Pandas, NumPy, SciPy/Scikit-learn, Matplotlib) and MATLAB, integrating multi-domain biomarker and autonomic nervous system metrics to characterize stress and recovery profiles.
- Authoring peer-reviewed manuscripts translating complex psychophysiological findings for scientific and applied health audiences.

Postdoctoral Research Fellow: Occupational Sleep Medicine

July 2024 – Sept 2025

Washington State University, Spokane, WA

- Developed and implemented Python-based signal processing algorithms to quantify sleep regularity (Sleep Regularity Index) from wearable and actigraphy data. Manuscript currently under peer review.
- Conducted qualitative data analysis (focus groups and interviews) on 22 active triathletes examining sleep and recovery behaviors using Qualtrics-based survey instruments; secured \$1,400 in grant funding supporting this research; manuscript currently under peer review and findings presented as an oral presentation at a national academic conference.
- Conducted qualitative data analysis (focus groups and interviews) on aviation personnel using thematic coding methodologies (ATLAS.ti); produced internal technical reports and evidence-based fatigue countermeasure guidance for an aviation industry funding partner.
- Synthesized peer-reviewed literature and developed technical reports and educational resources translating complex sleep science for regulatory and non-specialist audiences.

Graduate Research/ Teaching Assistant

Aug 2019 – June 2024

Washington State University, Spokane, WA

- Designed and executed a longitudinal human subjects study across 21 participants over 6 weeks — including high-frequency lab-based protocols (up to 6 visits/week) — encompassing pulmonary function testing, body composition, metabolic exercise testing, and biomarker analysis; performed statistical modeling, data wrangling, and data visualization using Python (Pandas, NumPy, Pingouin, SciPy, Seaborn, Matplotlib), Excel, and SPSS to derive and communicate research findings.
- Secured \$5,850 in competitive grant funding to support human-based proteomics analysis identifying ~400 unique proteins and 38 novel inflammatory biomarkers associated with acute-phase inflammatory response; presented findings via poster presentation at two academic conferences.
- Performed electronic data capture (EDC) and longitudinal data management using REDCap; contributed findings to peer-reviewed manuscripts and technical reports in collaboration with clinical and academic partners.
- Authored regulatory-compliant research documentation including IRB protocols, informed consent forms, and SOPs in accordance with institutional and federal requirements.

Athletic Trainer and Instructor of Health & Wellness

Aug 2016 – June 2019

Mount St. Joseph University, Cincinnati, OH

- Maintained clinical documentation using electronic health records (EHR) and standardized medical coding systems (ICD-10, CPT); curated patient progress notes and treatment plans aligned with healthcare industry standards.
- Collaborated with physicians and allied health professionals to design and implement evidence-based care strategies for athlete-patients, ensuring continuity of care and data-informed clinical decision-making.

Athletic Development Specialist July 2015 – July 2016

IMPACT Sports Performance, Buffalo, NY

- Designed and implemented individualized evidence-based performance and injury prevention programs for elite athletes, integrating data from assessments to guide programming decisions.
- Established and coordinated medical referral workflows and documented patient care in collaboration with hospital-based medical networks and sports medicine teams.

Graduate Assistant Athletic Trainer June 2014 – June 2015

University of Virginia, Charlottesville, VA

- Managed clinical patient care responsibilities—including injury assessment, rehabilitation, and referral coordination—while completing a master's degree in athletic training and contributing to one peer-reviewed publication.
- Developed and delivered evidence-based educational materials to healthcare teams through clinical in-services and journal clubs.

Skills:

Data Analysis & Programming: Python (Jupyter Notebook, VS Code; Pandas, NumPy, SciPy, Scikit-learn, Statsmodels, Pingouin, Matplotlib, Seaborn) | SQL/ MySQL | MATLAB | Excel | SPSS | Statistical Modeling | Data Wrangling | Data Visualization

Physiological & Domain Expertise: Sleep Physiology | Heart Rate Variability (HRV) | Psychophysiology | Allostatic Load | Exercise Physiology | Wearable & Actigraphy Data | Biomarker Analysis

Research Skills: Human Subjects Research | Study Design | Project Management | SOP/ IRB Protocol Development | Qualitative Data Analysis | Thematic Coding | Grant Writing (\$7,250 secured)

Scientific Communication: Scientific Writing & Medical Writing (AMA, APA) | Literature Review & Synthesis | Peer-reviewed Manuscripts | Technical Reports | Content Creation | Proof Reading | Database Navigation (PubMed, Cochrane, NIH) | Conference Presentations (Oral and Poster)

Tools & Platforms: Git/GitHub | MySQL Workbench | Google Workspace | Microsoft Office Suite (Word, PowerPoint, Excel) | ATLAS.ti | REDCap | Qualtrics

Education:

Postdoctoral Research Fellowship: Occupational Sleep Medicine, Washington State University	Sept 2025
Doctor of Philosophy: Nutrition & Exercise Physiology, Washington State University	June 2024
Master of Education: Athletic Training, University of Virginia	June 2015
Bachelor of Science: Athletic Training, Northern Kentucky University	Dec 2013

Affiliations:

National Provider Number (NPI)	#1003207531
Certified Strength and Conditioning Specialist (NSCA-CSCS)	#7248216927
Certified Athletic Trainer (NATABOC-ATC)	#2000015851
Sleep Research Society	
American College of Sports Medicine	
National Athletic Trainers' Association	
National Strength & Conditioning Association	
Sigma Phi Epsilon Fraternity Alumna (KY-Eta '13)	